

## Jing Yuan, PhD

Address: 715 P Street, Sacramento, CA 95814

Work: 916.326.9042

Email: [jing.yuan@water.ca.gov](mailto:jing.yuan@water.ca.gov)



## EDUCATION

**Ph.D., Interdisciplinary Ecology, University of Florida, 2015**

**Concentration: Geographic Information Systems**

Dissertation: *Metrics of Pattern Loss and Ecosystem Change in the Ridge and Slough Mosaic of the Everglades* (Advisor: Matthew J. Cohen)

**M.Sc., Natural Resources Conservation, Beijing Forestry University, 2009**

**B.Sc., Environmental Science, Southwest Forestry University, 2005**

## WORK EXPERIENCE

**Senior Environmental Scientist**, Water Use and Efficiency Branch, California Department of Water Resources, 07/2022 – Current

**Research Associate**, Center for Global Change & Earth Observations, Michigan State University, 07/2020 - 06/2022

**Research Associate**, National Center for Geographic Information and Analysis, University of Maine, 05/2018 - 06/2020

**Research Scientist**, Nature Conservation Academy, Qiaonv Foundation, 11/2017-04/2018

**Researcher**, Chinese Academy of Environmental Planning, Ministry of Ecology and Environment of China, 12/2015 - 10/2017

**Environmental Specialist (Intern)**, Great Tumen Initiative Secretariat, United Nations Development Programme, 09/2015 - 11/2015

**Research Assistant**, Ecohydrology Lab, University of Florida, 08/2009 - 08/2015

## PUBLICATIONS

### ***Published***

1. Chen, J., John, R., Kolluru, V., Mack, E. A., Fan, P., **Yuan, J.**, ... & Gutman, G. (2024). Natural assets, changes, and variations of the socioeconomic-environmental systems along the Asian drylands belt. *Environmental Research: Ecology*, 3(4), 045005.
2. **Yuan, J.**, & Chen, J. (2024). Disproportionate contributions of land cover and changes to ecosystem functions in Kazakhstan and Mongolia. *Scientific Reports*, 14(1), 21922.
3. Sciusco, P., Chen, J., Giannico, V., Abraha, M., Lei, C., Shirkey, G., **Yuan, J.**, Robertson, G. P., & Kellogg, W. K. (2022). Albedo-Induced Global Warming Impact at Multiple Temporal Scales within an Upper Midwest USA Watershed. *Land* 2022, Vol. 11, Page 283, 11(2), 283. <https://doi.org/10.3390/LAND11020283>

4. **Yuan, J.**, Chen, J., Sciusco, P., Kolluru, V., Saraf, S., John, R., & Ochirbat, B. (2022). Land use hotspots of the two largest landlocked countries: Kazakhstan and Mongolia. *Remote Sensing*, 14(8), 1805.
5. Venkatesh, K., John, R., Chen, J., Jarchow, M., Amirkhiz, R. G., Giannico, V., Saraf, S., Jain, K., Kussainova, M., & **Yuan, J.** (2022). Untangling the impacts of socioeconomic and climatic changes on vegetation greenness and productivity in Kazakhstan. *Environmental Research Letters*, 17(9), 095007.
6. Sciusco, P., Chen, J., Giannico, V., Abraha, M., Lei, C., Shirkey, G., **Yuan, J.**, Robertson, G. P., & Kellogg, W. K. (2022). Albedo-Induced Global Warming Impact at Multiple Temporal Scales within an Upper Midwest USA Watershed. *Land* 2022, Vol. 11, Page 283, 11(2), 283.  
<https://doi.org/10.3390/LAND11020283>
7. Chen, J., John, R., **Yuan, J.**, Mack, E. A., Groisman, P., Allington, G., Wu, J. (Jingle), Fan, P., Beurs, K. M. de, Karnieli, A., Gutman, G., Kappas, M., Dong, G., Zhao, F., Ouyang, Z., Pearson, A. L., Şat, B., Graham, N. A., Shao, C., ... Qi, J. (2021). Sustainability challenges for the social-environmental systems across the Asian drylands belt. *Environmental Research Letters*.  
<https://doi.org/10.1088/1748-9326/AC472F>
8. Abraha, M., Chen, J., Hamilton, S. K., Sciusco, P., Lei, C., Shirkey, G., **Yuan, J.**, & Robertson, G. P. (2021). Albedo-induced global warming impact of Conservation Reserve Program grasslands converted to annual and perennial bioenergy crops. *Environmental Research Letters*, 16(8), 084059.  
<https://doi.org/10.1088/1748-9326/AC1815>
9. **Yuan, J.**, Beard, K., & Johnson, T. R. (2021). A quantitative assessment of spatial patterns of socio-demographic change in coastal Maine: one process or many? *Applied Geography*, 134, 102502.  
<https://doi.org/https://doi.org/10.1016/j.apgeog.2021.102502>
10. Beard, K., Kimble, M., **Yuan, J.**, Evans, K. S., Liu, W., Brady, D., & Moore, S. (2020). A Method for Heterogeneous Spatio-Temporal Data Integration in Support of Marine Aquaculture Site Selection. *Journal of Marine Science and Engineering*, 8(2), 96.
11. **Yuan, J.**, & Cohen, M. J. (2020). Remote detection of ecosystem degradation in the everglades ridge-slough landscape. *Remote Sensing of Environment*, 247, 111917.
12. Johnson, T.R.; Beard, K.; Brady, D.C.; Byron, C.J.; Cleaver, C.; Duffy, K.; Keeney, N.; Kimble, M.; Miller, M.; Moeykens, S.; **Yuan, J.**; et al. A Social-Ecological System Framework for Marine Aquaculture Research. *Sustainability* **2019**, 11, 2522. <https://doi.org/10.3390/su11092522>
13. **Yuan, J.**, & Cohen, M. J. (2017). Spatial metrics for detecting ecosystem degradation in the ridge-slough patterned landscape. *Ecological Indicators*, 74, 427–440.
14. Ma, G., Yu, F., Wang, J., Zhou, X., **Yuan, J.**, Mou, X., Zhou, Y., Yang, W., & Peng, F. (2017). Measuring gross ecosystem product (GEP) of 2015 for terrestrial ecosystems in China. *China Environmental Science*.
15. Qi, J., Zheng, B., Li, M., Yu, F., Chen, C., Liu, F., Zhou, X., **Yuan, J.**, Zhang, Q., & He, K. (2017). A high-resolution air pollutants emission inventory in 2013 for the Beijing-Tianjin-Hebei region, China. *Atmospheric Environment*, 170, 156–168.  
<https://doi.org/https://doi.org/10.1016/j.atmosenv.2017.09.039>
16. **Yuan, J.**, Cohen, M. J., Kaplan, D. A., Acharya, S., Larsen, L. G., & Nungesser, M. K. (2015). Linking metrics of landscape pattern to hydrological process in a lotic wetland. *Landscape Ecology*.  
<https://doi.org/10.1007/s10980-015-0219-z>

17. **Yuan, J.**, Cui, G. F., & Lei, T. (2009). Major hydro environmental factors affecting the community composition of wetland submerged plants in Beijing. *Chinese Journal of Ecology*, 28, 2189–2196.
18. Zong, X., Cui, G., & **Yuan, J.** (2008). Contingent valuation of the existent economic of giant panda (*Ailuropoda melanoleuca*). *Acta Ecol Sin*, 28, 2090–2098.

### ***In Review/In Preparation***

1. Yuan, J. Conrad, L. Andrew L. AI-Driven Publication Inventory and Collaboration Mapping in California Water Resources Management (In prep.)
2. Yuan, J., Harvey, J., Jones, J. and Choi, J., Remote sensing of vegetation biomass to predict changing flow resistance for improved hydrological modeling (In prep.)

### **GRANTS**

NSF PIRE: Integrating traditional knowledge, translational science, and emerging technology for adaptive pastoral systems in Mongolia and Kazakhstan. Chen, J., John, R., Knight, C., Hartman, D., Fan, P., **Yuan, J.**, Kapps, M., Furst, C., Schemidt, J.G., Meiramova, S., Amartuvshin, A., Baltashevich M., Karnieli, A., Aryngaziye, Batkhishig, O., Hoshino, B., Kussainova, M., Lkhagvaa D., Davaanyam T., & Spaeth K., Submitted to Partnerships for International Research and Education, NSF for \$1,500,000, *Not funded*.

NASA LULCC: Urbanization and sustainability under global change and transitional economies: synthesis from southeast Asia. Fan, P., Cho, M., Lin Z., Ouyang, Z., Qi, J., Chen, J., Moran F. E., & **Yuan, J.** Submitted to the National Aeronautics and Space Administration, Land-Cover and Land-Use Change Program (LULCC) for \$3,000,000. *Not funded*.

### **PRESENTATIONS (most recent 5 years)**

#### ***Oral Presentations***

- **Yuan, J.** Leveraging Google Earth Engine for Efficient Land Use Mapping and Water Use Estimation, 2024 DWR Environmental Science Workshop, November 14-15, 2024
- **Yuan, J.** Unveiling Three Decades of Land Use Hotspots (1990–2020): Analyzing Kazakhstan and Mongolia via Cloud Computing with Google Earth Engine, CA Silver Jackets: Flood Risk Management Technical Workshops, DWR Joint Operations Center, January 17-18, 2024.
- Mubako, S., **Yuan, J.**, Vollmer, J., Mostafavi, M., Hayes, S., California Land Use Data Through the Years: From Hand Drawn Maps to Remote Sensing and Public Distribution, Tahoe City, CA, 2023 DWR Environmental Science Workshop, October 11-12, 2023.
- **Yuan, J.**, Chen, J., Disproportionate contributions of land cover and changes to ecosystem functions in Kazakhstan and Mongolia, ESA annual meeting, Portland OR, August 6-11, 2023.
- **Yuan, J.**, Harvey, J., Choi, J. and Jones, J., Remote sensing of vegetation biomass to predict changing flow resistance for improved hydrological modeling, Greater Everglades Ecosystem Restoration (GEER 2023), Coral Spring, Florida, April 17-20, 2023.

- **Yuan, J.**, Chen, J., Sciusco, P., Kolluru, V., Saraf, S., John, R., & Ochirbat, B. Land Use Hotspots of the Two Largest Landlocked Countries: Kazakhstan and Mongolia, Japan Geoscience Union Meeting, Chiba, Japan, May 22-27, 2022. (Online)
- **Yuan, J.**, Chen, JQ, John R., Knight, C., Mediating roles of Land use and Land cover change on ecological functions and socioeconomic shifts in Mongolia and Kazakhstan. V International Agritech Summit: Development of Agro-Industrial Complex through Research, Education, and Innovative Production. Almaty, Kazakhstan, December 2020. (invited)

### ***Poster Presentations***

- **Yuan, J.**, Chen J., Disproportionate contributions of land cover and changes to ecosystem functions in Kazakhstan and Mongolia, AGU Fall Meeting, San Francisco CA, Dec. 11-15, 2023.

### **TEACHING AND MENTORING**

**Dissertation committee**, Department of Ecology and Environmental Sciences, University of Maine, Spring 2023 - Current

- Nathan Smith, Dissertation “Social-ecological system analysis of recirculating aquaculture development”

**Guest lecturer**, SIE509 Principles of Geographic Information Systems, Fall 2019  
School of Computing and Information Science, University of Maine

- Guest lecture on American Census and American Community Survey

**Mentor**, Student Science Training Program, Summer 2015

Center for Precollegiate Education and Training, University of Florida

- Mentored one high school student in field data collection and geospatial analysis

### **PROFESSIONAL SERVICES**

**Environmental Science Workshop Planning Committee, Department of Water Resources, 2024**

- Conceptualized workshop theme as data science and invited speaker selections.
- Contributed insights and recommendations on workshop content, ensuring alignment with departmental objectives and current environmental challenges.

### **Grant Reviewer**

- NOAA Small Business Innovation Research (SBIR)

**Manuscript Peer Reviewer** for 6 journals and 50+times:

- Sustainability, 30+ manuscripts/times
- Remote Sensing, 20+ manuscripts/times
- International Journal of Environmental Research and Public Health, 2 manuscripts/times
- Ecological Processes, 2 manuscripts/times
- Ecological Applications, 1 manuscript/times
- Watershed Ecology and Environment, 2 manuscripts/times

- Wetland, 1 manuscripts/times

**Secretary**, Sino-Ecologists Association Overseas, 2018-2020, 2020-2020

### **PROFESSIONAL MEMBERSHIPS**

- American Association of Geographers (AAG) (2019-current)
- American Geophysical Union (AGU)(2013-Current)
- International Association of Landscape Ecology (IALE) (2015-Current)
- Ecological Society of America (ESA)(2010-Current)
- Society of Wetland Scientist (SWS)(2010-2015)

### **AWARDS AND HONORS**

- Sino-Eco Outstanding Service Award, Sino-Ecologists Association Overseas, 2022
- China Scholarship, Chinese Ministry of Education, 2009-2013
- Outstanding Graduate Fellowship, Beijing Forestry University, 2006-2009

### **CERTIFICATES**

- Open Water Scuba Dive, Diver #15050T2951, PADI, 05/2015
- University Teaching, The University of Hong Kong and offered through Coursera, 10/2023
- Emerging Leader Cohort, CA Department of Water Resources, 8/2025

### **TECHNICAL SKILLS**

| Technical Area                  | Product             | Proficiency |
|---------------------------------|---------------------|-------------|
| Remote Sensing/Image Processing | Google Earth Engine | Expert      |
|                                 | ENVI                | Expert      |
|                                 | IDRISI              | Expert      |
|                                 | Erdas Imagine       | Expert      |
| GIS                             | ESRI ArcMap         | Expert      |
|                                 | ESRI ArcMap Pro     | Expert      |
|                                 | QGIS                | Good        |
| Database                        | PostgreSQL/PostGIS  | Good        |
| Operating Systems               | Linux               | Good        |
| Programming Languages           | R                   | Expert      |
|                                 | JavaScript          | Very Good   |
|                                 | HTML                | Good        |
| Web Development                 | Adobe Dreamweaver   | Good        |
| Video Editing                   | Adobe Premiere Pro  | Good        |
|                                 | Camtasia            | Good        |